

DR MUHAMMAD SHAHID QURESHI

Educational Qualifications:

Ph.D. Institute of Space & Planetary Astrophysics, University of Karachi, Thesis entitled ON THE COMPARATIVE STUDY OF MATHEMATICAL MODELS FOR EARLIEST VISIBILITY OF THE LUNAR CRESCENT AND THEIR MODIFICATION, (2008).

M.Phil. Department of Mathematics, University of Karachi, Thesis entitled "Super symmetric Quantum Mechanics"(1995)

M.Sc. Applied Mathematics, Department of Mathematics, University of Karachi, securing First Class with Distinction (75.1%) marks, specialisation in ASTRONOMY & QUANTUM MECHANICS(1983)

B.Sc.(Hons) Applied Mathematics, Department of Mathematics, University of Karachi, Securing First Class with Distinction(75%) (1980)

G.R.E. General (with score over 1500) and Subject (score of 750)(1994)

Computer Related:

During on job training at U.B.L. Computers Division, worked as Computer Programmer and as System Analyst, Programming in ASSEMBLY & COBOL

During one semester stay at C.N.S., Nilore, learnt Programming in FORTRAN

Working knowledge of QBASIC, C++, DBASE, MS OFFICE, CHIWRITER, WINDOWS.

Professional & Teaching Experience

PROFESSOR, in the Department of Mathematical Sciences, Faculty of Computer Science, Institute of Business Administration, Karachi, June 1, 2011 to-date.

ASSISTANT PROFESSOR, in the Department of Mathematical Sciences, Faculty of Computer Science, Institute of Business Administration, Karachi, June 30, 2010 to May 31, 2011.

ASSISTANT PROFESSOR, Institute of Space & Planetary Astrophysics, University of Karachi, July 2004 to June 29, 2010.

ASSISTANT PROFESSOR, in the Department of Mathematics, University of Karachi, Feb. 7, 2002 to July 2004.

ASSISTANT PROFESSOR, in the Department of Computer Science, University of Karachi, June 1, 2000 to Feb. 6, 2002.

ASSISTANT PROFESSOR, in the Department of Mathematics, University of Karachi, May 1, 1999 to May 30, 2000.

ASSISTANT PROFESSOR, in Mohammad Ali Jinnah University, Karachi, January 1, 1999 to April 30, 1999 (Taught both Mathematics and Computer courses).

ASSISTANT PROFESSOR, in the Department of Mathematics, University of Karachi, Aug. 15, 1995 to December 31, 1998.

LECTURER, in the Department of Mathematics, University of Karachi, Sept. 22, 1984 to Aug. 14, 1995.

COMPUTER PROGRAMMER, in U.B.L., Computers Division, Karachi, April 1980 to August 1982.

PART TIME APPOINTMENTS:

VISITING FACULTY, Institute of Business Administration, from August 2002 to December 2004, & August 2008 to June 2010.

VISITING FACULTY, BCCI FAST Institute of Computer Science, from January 1993 to December 1997.

VISITING FACULTY, Hamdard University, from June 1999 to December 1999.

VISITING FACULTY, Iqra University, from August 2003 to December 2006.

VISITING FACULTY, Sir Syed University of Engineering, from August 1995 to December 1997.

VISITING FACULTY, in the Evening Program MIBM, of the Department of Mathematics, University of Karachi, from 1997 to 2002.

VISITING FACULTY, in the Department of Computer Science University of Karachi, January 1997 to December 1998 & March 2002 to May 31, 2010.

VISITING FACULTY, in the Bachelor of Chemical Technology program of the Department of Applied Chemistry, 1995 to 2000.

VISITING FACULTY, in the Department of Food Science & Technology, July 2000 to Dec. 2004.

CONSULTANT ASTRONOMER, P.I.A. Planetarium, Karachi, Nov. 1991 to Aug. 1992.

Administrative Experience:

CHAIRPERSON, Department of Mathematical Sciences, Faculty of Computer Science, Institute of Business Administration, Karachi, July 2010 to-date.

IN CHARGE, Institute of Space & Planetary Astrophysics (ISPA), University of Karachi, September 17, 2003 to June 29, 2010.

CO-ORDINATOR, BS Evening program, in the Department of Computer Science, University of Karachi, July 16, 2001 to Feb. 6, 2002.

CONVENOR, Campus Committee for Residential Area, University of Karachi, An administrative position, held from November 1997 to June 1998.

IN-CHARGE OBSERVATORY, The Karachi University Observatory, December 1984 to June 1993.

Research Publications:

A Correlation Between Lunar Distances & Earthquake Frequencies, Journal Geography and Geology, Vol. 3, No. 1, 171-177, Canada, Sept. 2011.

Thunderstorm and Rainfall Frequencies over Pakistan, Elixir Space Sci. 37, 3981-3985, August 2011.

IBook: Interactive and semantic Multimedia Content Generation for eLearning, IJCSI International Journal of Computer Science Issues, Volume 8, Issue 3, May 2011.

Comparative Study of VoIP over WiMax and WiFi, IJCSI International Journal of Computer Science Issues, Volume 8, Issue 3, May 2011.

Climatic Variability and Linear Trend Models for Major Cities of Pakistan , Journal of Geography and Geology, Vol. 2, No.1, pp. 83-92, September 2010.

A New Criterion of Earliest Visibility of New Lunar Crescent, Sindh Univ. Res. Jour. (Sci. Sec.), 42(1), 1-18, 2010.

On Origin Theories of Ring Galaxies: Observational Data and Modelling, Astronomy & Applied Mathematics, Vol. I, Ed. S Nuritdinov, Pub.: Fed. Urdu University, Karachi, 67-79, 2009, ISBN-978-969-9407.

Path of Best Visibility of New Lunar Crescent, Astronomy & Applied Mathematics, Vol. I, Ed. S Nuritdinov, Pub.: Fed Urdu University, Karachi, 156-161, 2009, ISBN-978-969-9407.

Celestial Objects Visibility Concern in Karachi, Astronomy & Applied Mathematics, Vol. I, Ed. S Nuritdinov, Pub.: Fed Urdu University, Karachi, 185-194, 2009, ISBN-978-969-9407.

Error Minimization in Polynomial Approximation of ΔT , J. Astrphys. Astr., 29, 363-355, 2008.

Assessing Polynomial Approximation of ΔT , J. basic appl. Sci., 4(1), 1-4, 2007.

Methods of Calculating Crescent Lengths, J. basic appl. Sci., 3(2), 62-75, 2007.

"Taqweemat aur Qudrati Mazahir, Al-Tafseer, 2(1), 101-107, 2006.

Interaction Studies of Omeprazole with Mefloquine, Pyrimethamine and Sulfadoxine, Pak. J. Pharm. Sci., 19(4), 314-321, 2006

A Comparative Study of Arithmetic Lunar Calendar, Observational and Predicted Lunar Calendar for Pakistan for Years 2000-2004, J. basic appl. Sci., 2(2), 91-103, 2006.

A New Variable Step Size Algorithm for Solving Initial Value Problems, J. basic appl. Sci., 2(1), 37-44, 2006.

On the Modern Approach to the earliest Visibility of Lunar Crescent, Kar. Un. J. Isl. St., VII(I-II), 54-76, 2005.

A Fuzzy Mathematical Model of Maximum Likelihood Estimates, J. basic appl. Sci., 1(2), 95-100, 2005.

Differential Mental Growth Model, J. basic appl. Sci. 1, 11-13, 2005.

From No-Go Theorems to Supersymmetry Algebra, Kar. Univ. J. Sc., 32(1&2), 55-62, 2004.

SUSY Algebra for One Dimensional Quantum Mechanical Systems and its Application, Kar. Univ. J. Sc. 28(1),83-95, 2000.

Generalisation of Operator Method and its Application, Kar. Univ. J. Sc. 28(1), 83-95, 2000.

Solvable Supersymmetric Systems", Kar. Univ. J. Sc. 23(1&2),27-38, 1995.

Other Publications

Co-Edited the Book Astrophysics and Applied Mathematics, Vol. I, 2009, Published by Federal Urdu University of Arts Science and Technology, Karachi, in celebration of the World Space Year 2009.

Supervised Research

Currently supervising:

MPhil thesis on Studies on the slowing rate of the axial rotation of the earth, at Institute of Space & Planetary Astrophysics, University of Karachi. Oral defence held, Referee reports received, waiting for award of degree.

MPhil thesis work on Mathematical Modelling of the Visibility on the Night Sky, Institute of Space & Planetary Astrophysics, University of Karachi. Submitted in August 2011.

PhD dissertation on Origin Theory of Ring-Like Self-Gravitational Structures: Development on the Basis of Observational Data and Mathematical Modelling, at Institute of Space & Planetary Astrophysics, University of Karachi. Submitted in July 2011.

MPhil thesis work on Observational Lunar Calendar for Higher Latitudes, at Institute of Space & Planetary Astrophysics, University of Karachi. Work in progress.

MPhil thesis work on A Study of the Calendrical Aspect of the Celestial Phenomena, Institute of Space & Planetary Astrophysics, University of Karachi, Work in progress.

Supervised Dissertations:

Teachers Evaluation Using Genetic Algorithm, BS Thesis, 2001, Mr. Shah Jamal Alam, Department of Computer Science, University of Karachi.

Unified Modelling Language for Java, BS Thesis, 2001, Mr. Ishtiaq Ahmed, Department of Computer Science, University of Karachi.

Design Pattern Analysis, BS Thesis, 2001, Mr. Tahir Mansoori, Department of Computer Science, University of Karachi.

"Image Enlargement using Iterated Function System, BS Thesis, 2002, Ms. Nadia Alvi. Department of Computer Science, University of Karachi.

Network Intrusion Signature and Analysis, BS Thesis, 2002, Ms. Sabaina Bukhari, Department of Computer Science, University of Karachi.

"Application of Object Oriented Design Patterns in Compiler Construction, BS Thesis, 2002, Ms. Sarah Sameen, Department of Computer Science, University of Karachi.

Complexity Based Study on Object Oriented Metrics in View of Artificial Neural Networks, BS Thesis, 2002, Department of Computer Science, University of Karachi

Conferences & Seminars (Attended)

Attended as a research student the "International Conference on Relativity & Superstrings" held at Quaid-e-Azam University, Islamabad, Feb. 1989.

Attended a workshop on Elementary Particles & Early Universe Held at CASM, LUMS, in April 2005.

Attended a workshop on Computational Number Theory & Cryptography Held at CASM, LUMS, in April 2005.

Talks Delivered

"Supersymmetric Quantum Mechanics" in "Einstein Seminar-1994" held at SPARCENT, Karachi in April 1994.

"Supersymmetry Algebra" in a series of Seminars held at the Department of Mathematics, University of Karachi, in Oct. 1994.

"Calculation Involved in the Study of Solar Eclipse" in a seminar on "SOLAR ECLIPSE-1995" held at the Department of Physics, University of Karachi, Nov. 1995.

Theory of Fractals and its scope in view of Graphic Designs, in a series of Seminars held at the Department of Computer Science, University of Karachi, August 2001.

Earliest Crescent Visibility: Ancient, Medieval & Modern Approaches in a Seminar Held in Federal Urdu University, Karachi, on the occasion of World Space Week, 2007.

Delivered a talk on Earliest Visibility of New Lunar Crescent as Invited Speaker in Pakistan Nuclear Regulatory Authority, Islamabad, April 8, 2010.

Papers Presented

Presented a paper on Earliest Visibility of Lunar Crescent in the Winter Conference on Mathematics held in Dec. 2004 at the Centre of Advanced Studies in Lahore, University of Management Sciences.

Delivered a talk on Ruet-i-Hilal (Crescent Sighting), Observational Aspects at a seminar held at the Department of Mathematical Sciences, Federal Urdu University, Karachi, June, 2005.

Presented a paper on Dynamics of Moon and the Challenges of Earliest Visibility of Lunar Crescent, in the Seminar on Space Science & Challenges of 21st Century, held on November 21, 2005, at University of Karachi, in collaboration of SUPARCO in connection with the World Space Week 2005.

Presented a paper on Comparative Study of Arithmetic, Computed and Observational Lunar Calendars in the Summer Conference on Mathematics held in July 2005 at the Centre of Advanced Studies in Lahore University of Management Sciences.

Presented a paper on Length of Lunar Crescent in the Conference on Mathematics held in February 2006 at the Centre of Advanced Studies in Lahore University of Management Sciences.

Presented a paper on Earliest Visibility of New Lunar Crescent in the Word Space Seminar, held in University of Karachi in connection with the World Space Week-2008, November, 2008.

Presented a paper on Physics of Earliest Crescent Visibility in the International Conference on Physics & the World of Today, Held in the Department of Physics, University of Karachi, Karachi, Dec. 2008.

Presented a paper on Invisibility Period of Lunar Crescent in the 35th All Pakistan Science Conference, held in University of Karachi, Dec 20-23, 2008.

Presented a paper on Lunar Crescent: From Ancient to the Modern Era in the First National Conference on Mathematical Sciences, held in the Department of Mathematical Sciences, University of Karachi, Karachi, Mar. 22, 2010.

Extra-curricular Activities

ELECTED MEMBER, ACADEMIC COUNCIL, University of Karachi from Nov. 1993 to Nov. 1996, and from August 2005 till June 2010.

HONARARY EDITOR, (since 1992) "Science Digest", Karachi.

Published a number of articles related to Astronomy in the daily "Jang", the daily Dawn and "Science Digest".

Remained an active cricketer throughout student life, represented teams of School(1972-73), Adamjee Science College(1974-75), Karachi University Faculties (1976-77), Centre for Nuclear Studies(1984) and various local clubs in a number of tournaments of Karachi Cricket Association.

Served as "Elected Joint Secretary" of "EPSILON CIRCLE" the departmental student society(1978-79) of the Department of Mathematics, University of Karachi. Served as Elected Joint Secretary of the "Karachi University Students Union"(1979-80).